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INST414
Sprint 3
QSSR track

Modeling Strategy & Implementation

The statistical approach I will use is linear regression as my research question revolves around finding relationships between social media hours and topics like reported mental health and distraction. Linear regression is effective here as it would enable us to see if the relationship between social media/screentime and mental health score is positive, negative or neutral in addition to seeing how strong the relationship is. While using this model, major assumptions include that all reports were made independent of each other and that our variables will follow a linear pattern.

The parameters I chose for the model were a test size of 0.2 which indicates that the data was split in a 80/20% ratio for training and testing. It is often the standard parameter, I also chose it as the dataset feels fairly small (481 entries) which would lead to our subgroups for social media usage also being small. It is noticeable that there are gaps in the size of subgroups to consider. Certain groups such as ‘less than an hour’ have notably less reports than others which could influence overall accuracy.

Less than an Hour	Between 1 and 2 hours	Between 2 and 3 hours	Between 3 and 4 hours	Between 4 and 5 hours	ore than 5 hours
34	70	101	93	67	116

Model Development & Training

a) The baseline model focuses on predicting depression rate based on social media usage hours. After converting the social media usage column to numerical representations, a linear regression model was built and retuned a mse of 1.5197, R squared of 0.0757, mae of 1.0461 and a slope of 0.44

b) Linear regression model with a single independent variable (average social media hours) and a single predictor variable (depression rate). The dataset was split into 80% for training and 20% for testing. After this, a linear model was created and fit onto the training set. Using this, predictions were made as well as calculations for the R squared, mean squared error and mean absolute error.

The following model was also a linear regression but followed average social media hours as the independent variable while having distraction_when_busy_score , sleep_issue_rate and difficulty_concentrating_score as dependent variables which were used as a proxy for distraction.

c)

	Mean Squared Error	R-squared	Mean Absolute Error:	Coefficient	
Depression/ down model.	1.5197	0.0757	1.0461	0.4416	
Distraction model	1.8006	0.1127	1.1518	0.4801, 0.1983, 0.4391	

d) The main form of feature engineering present was the altering of features to meet certain types, the average social media hours column was converted from being categorical representations (“less than 1 hour”, “more than 5 hours”) to numeric codings which allowed the model to function since this was set as the independent variable.

Model Evaluation & Diagnostics

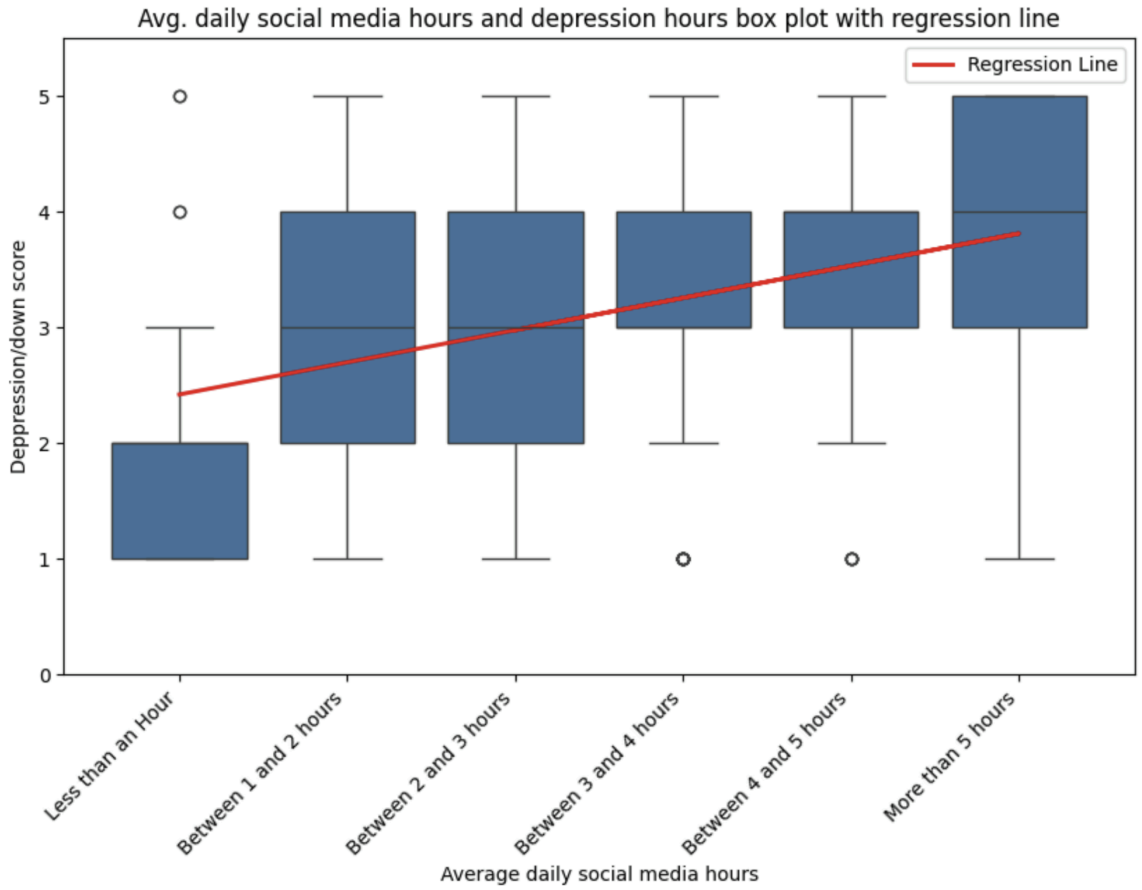
The hypothesized effect (social media/screentime use impacts mental health/distraction) seems to be true as models bring back positive coefficients (0.4416 model 1, avg. 3.725 model 2) which also represents the slope indicating a positive correlation/relationship between social media hours and mental health/distraction. The value 0.4416 coefficient would indicate that for every shift in social media hour group category, reported average increases by 0.4416. To translate this to our data set, around every 2 shifts in daily social media usage hours levels results in reported average depressed/down rates which is notable as the scale goes from 1 to 5. The same would apply to the coefficient 0.4801,0.1983,0.4391 for distraction when busy score, sleep issues rate and difficulty concentrating scores. The average daily social media hours seems to have the strongest correlation on the distraction when busy score and depressed/down rates which are the main target in the goal of this project.

Results & Interpretation

a) Primary results: Our models help push that there's a relationship between social media usage and mental health/distraction. There are identified positive relationships in our models between the average social media hours and reported depressed/down scores as well as our proxies for distraction (distractability when busy score, sleep issues ,difficulty concentrating).

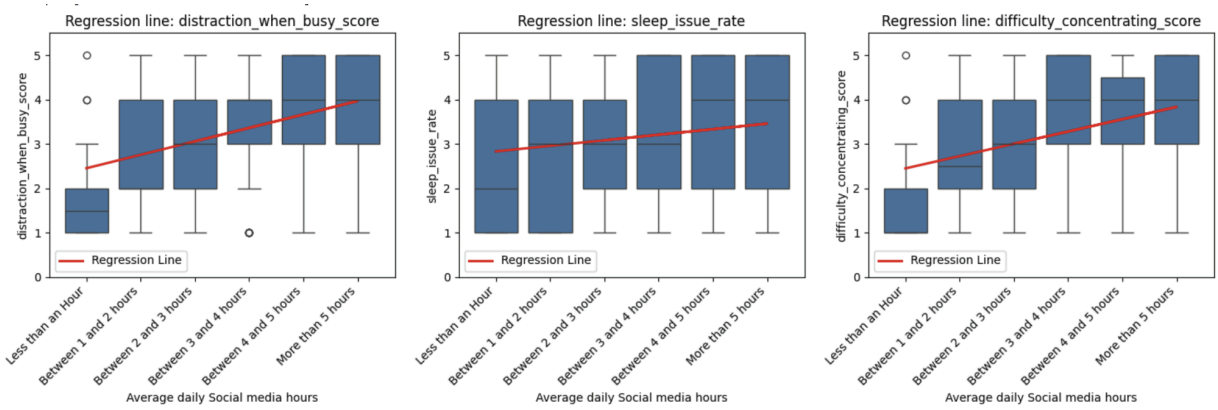
   This is visible in our coefficients which represent a slope in this case, 0.4416 for the depressed/down model and average 0.3725 for our distraction model. This indicates a positive slope showing that the score tends to increase as the social media hours increase. Individual running of ANOVA tests would also bring back a significant value indicating a relationship. This information is in line with my expectations from sprint 2 but I was surprised by the r squared values being lower than expected.

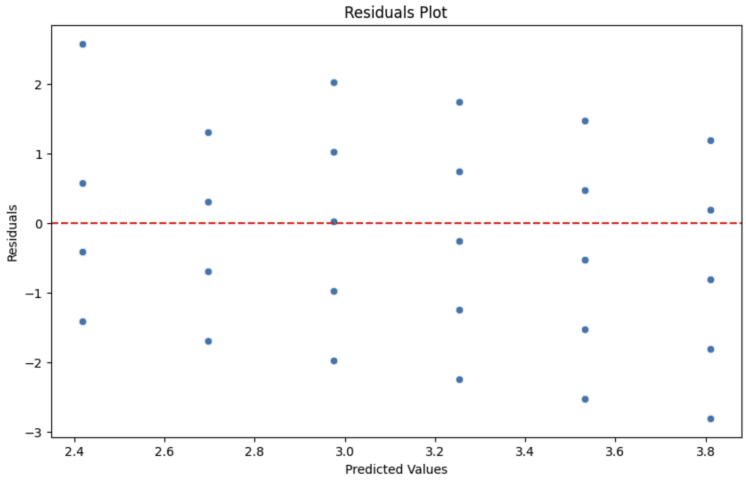
b) Box plot for depression/down model with regression line. Represents distribution of reported scores between average daily social media hours groups and shows the regression line which indicates a positive relationship.



Box plots for dependent features in a distraction model with a regression line. Visualizes the distribution relationship between average daily social media hours and respective dependent

features. Shows a positive relationship in all represented features.





c) The models give information that is useful to my problem, the coefficients (slope) being positive does indicate at least a positive relationship which is valuable to the research. The r squared values are also useful, although the r squared values are fairly low (0.0757 ,0.1127) which means our models are responsible for 7.5% and 11.2% of the variance, considering the context of the problem these values are very useful and indicative. The problem has many additional factors as there's a variety of things that could influence someone's mental health or distraction rate such as finance, habits, overall health etc. that aren't available in the data, this means that 0.11 and 0.07 r squared values can show that social media hours does somewhat of an impact. Overall the positive coefficient/slope value as well as r squared values are large enough to indicate that the hypothesis has some validity. The models don't tell us if there's causation as they currently only identify positive correlations, there isn't enough information in the data alone to conclude social

media usage is the direct causation without further research. The generalizability of the models may be challenged when faced with different demographics as it mainly deals with responses from people who are students and has a fairly low amount of data to work with.

d) There are a variety of studies relating to social media use hours and its influence on mental health or distractibility. A study on the association between time spent on social media and internalizing/externalizing problems among youth in the US (Riehm, K. E) Indicated that there was notable increase in mental health issues amongst people who made use of social media more than 3 hours a day. This falls in line with information from the model which indicates a positive correlation between social media use hours and increased reported depressed/down rate. The findings/trends from these models are consistent with a lot of information I've gathered prior about the influence of screen time/social media on individuals mental health and how it can function as a distraction.

Limitations, Assumptions & Threats to Validity

a) This approach had some limitations due to data. Since there's only 481 inputs, when we begin looking into different subgroups they sometimes end up having very low samples which can reduce reliability. Another limitation relates to how the initial survey was structured with categorical variables and using a 1-5 numeric scale which resulted in having to convert data in some areas to get certain sections to run and also added limits in the amount of ways data could be explored.

b) We cannot directly claim causality with this model because the problems this model was made to cover can be influenced by many factors outside of the scope of the data we have access to. For example there are many things that could influence a users reported depressed/down rate outside of screentime/social media use. The model has some potential risks of overfitting, particularly due to some groups having low counts which may make it difficult to get enough information that can be generalized to other sets.

c) The model may not generalize as well into different populations as the primary members where individuals surveyed at a university so a large portion of the data collected is students which may fail to take into account other circumstances if it is expanded to other populations

especially classes of people which may not pursue a college education. It may have a harder time generalizing to older and younger populations, younger people likely make use of social media more due to increased access to technology and the internet and older populations who didn't grow up with as much access might not have the same level of activity so age could be a potential factor.

e) I think my approach didn't function as well as initially expected but still gave back some useful data such as general positive trends between hours on social media factors such as the depression rate and distraction scores. One thing I would do differently is re-examine possible model options that may have been more accommodating to how the dataset was initially structured. Overall my problem statement is still fairly accurate as there does seem to be evidence that there's some kind of positive correlation/trend between social media hours and poor mental health/distractability.

Sprint 4 Plan & Refinement Strategy

a) In sprint 4 I could expand on exploring other areas for relationships such as occupation or relationship status and ultimately find areas of the models to update or expand on. I would also explore implementing regularizations and seeing if there's notable differences.
b) Core challenges during this sprint was deciding which model type would end supporting the mission of this project. Since the overall mission for my project is to find if there's a meaningful relationship, I ended up seeing a linear model as an option. Another challenge was considering how to handle the cleaned data set while implementing a model. I ended up having to do some conversions (categorical to numerical) in order to get my data to work with a linear model. Another challenge was finding appropriate approaches to model evaluation
c) My main goals for sprint 4 is updating the organization of the github files, applying appropriate model evaluation strategies for my implementation of linear regression. In

addition making use of the secondary dataset to further supplement findings would be useful. I should be able to properly present an analysis once those areas are fixed.

Self-Assessment

I feel confident in the results of the model as they indicate a positive relationship which is what I was initially expecting out of my research topic. I would give 8 out of 10 as there are some issues such as sample sizes in subgroups which I think reduces the validity of the models to an extent. One challenge has been handling additional information from the model as although it does bring back necessary information such as the slope and r squared, the way the data is structured makes additional graphs tedious. I also wanted to implement another type of model. Overall, getting the core information out of the model went better than expected but there's some areas such as testing different regularizations and validation tests that could be added or expanded on.

Citations

Riehm, K. E., Feder, K. A., Tormohlen, K. N., Crum, R. M., Young, A. S., Green, K. M., Pacek, L. R., La Flair, L. N., & Mojtabai, R. (2019, December 1). *Associations between time spent using social media and internalizing and externalizing problems among US youth*. JAMA psychiatry. https://pmc.ncbi.nlm.nih.gov/articles/PMC6739732/